

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (EVT) (CBCS) Degree Examinations

May – June 2025

22EVT231 – AUTOMOTIVE COMPUTER CONTROLLED SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Analyze the functioning of the digital ignition system.	06	L*3	1	3
b) Examine the role of Memory chips in automotive computers and compare their functionality with those used in personal computers.	08	L3	1	3
c) Analyze the working principle of a single-point fuel injection system.	06	L3	1	3
2. a) Examine the Key requirements for the effective functioning of computer controlled systems in automotive applications.	10	L3	1	3
b) Analyze the role of stability control in improving vehicle handling during understeer and oversteer conditions. Evaluate the significance of additional sensors in enhancing the effectiveness of stability control systems.	10	L3	1	3
3. a) Analyze the roles of the electronic vacuum regulator and the exhaust gas recirculation (EGR) Valve in engine Management systems.	10	L3	2	3
b) Explain the working of a four channel Anti – lock Breaking system (ABS) With a diagonal Split hydraulic Circuit.	06	L3	2	3
c) Analyze the fundamental parts of a computer and evaluate their interdependence in ensuring optimal system performance and functionality.	04	L3	2	3

Unit – II

4. a) Analyze the procedure for testing circuit continuity using a Multimeter. Evaluate the effectiveness Using a buzzer versus the ohm – meter scale in diagnosing (diagnosing) Electrical faults.	10	L3	3	3
b) Analyze the function of the throttle position switch in engine management. Evaluate how its role in adjusting the air-fuel ratio at idle and full throttle impacts engine performance and efficiency.	10	L3	3	3
5. a) Analyze the application of the potential divider principle in an air flow sensor.	08	L3	3	3
b) Analyze the working principles of solenoid-operated actuators in automotive systems. Compare the duty cycle method and pulse width modulation (PWM) in terms of control precision and efficiency.	08	L3	3	3
c) Analyze the role of fault analysis data accessed through the diagnostic link in modern vehicles.	04	L3	4	3

P.T.O.

6. a) Explain the working Principle of optical sensors in opto electronic devices. Evaluate the advantages and limitations of using photodiodes and phototransistors in automotive applications.
- b) Analyze the characteristics of the HT voltage trace for a single cylinder. Evaluate its significance in diagnosing ignition system performance and identifying potential faults.

10 L3 4 3

10 L3 4 3

Unit – III

7. a) Analyze the role of CAN communication protocols in modern vehicle systems. Evaluate their impact on real-time data exchange, fault diagnostics, and overall vehicle performance.
- b) Analyze the working principle of a Local Interconnect Network (LIN) for vehicle communication. Draw and explain the block diagrams of the master and slave nodes, evaluating their roles in ensuring efficient data transmission.
8. a) Analyze the Flex Ray communication topology used in automotive networks. Compare and contrast the multidrop topology and star topology in terms of data transmission efficiency, reliability, and fault tolerance.
- b) Analyze the GPS control configuration used in vehicle navigation systems. Evaluate its components and their roles in ensuring accurate positioning and real-time tracking.

10 L3 5 3

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10 L3 5 3

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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
